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On the antenna pattern of an orbiting interferometer

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ABSTRACT

The response of an interferometer changing its orientation with respect to a fixed reference frame is analysed in terms of the beam-pattern factors and the polarization-averaged antenna power pattern. Given the motion of the antenna, the latter quantity describes the directionality of the antenna as a function of time.

An interesting case is represented by the class of motions where the plane of the detector is constrained to move on the surface of a cone of constant aperture; at the same time, the two arms are rotating around a vertical axis. This picture describes, in particular, the motion of *LISA*, a proposed space-based laser interferometer, as well as of other planned missions. The overall sky coverage, and that of the Galactic plane in particular, is provided as a function of the aperture of the cone.

Similarly, one can consider the case of an Earth-based interferometer. Using the same formalism, one can derive a simple expression for the antenna pattern, averaged over the time of arrival of the signal, as a function of the position and orientation on the Earth's surface. In particular, there turn out to be particular values for the terrestrial latitude and the inclination angle with respect to the local parallel which render the time-averaged antenna response perfectly isotropic.

In the frequency domain, the general result is that the motion of the detector introduces in the instrumental response to a long-duration continuous signal a few harmonics of the orbital frequency, whose magnitude depends on the position of the source in the sky. In particular, we describe the response of *LISA* to sinusoidal waves coming from a few known binary systems in our Galaxy.

Key words: gravitation – instrumentation: detectors – instrumentation: interferometers.

1 INTRODUCTION

Gravitational waves in the low-frequency regime (10^{-4} to 10^{-1} Hz) can be observed only from space, due to terrestrial disturbances. In space, the only technique currently available, besides pulsar timing, is based on Doppler tracking of an interplanetary spacecraft (Eastabrook & Wahlquist 1975). However, this relatively inexpensive method has not provided enough sensitivity, thus far, for a detection. While better sensitivities may be expected in the near future, with advanced spacecraft such as *CASSINI*, much more ambitious projects for gravitational wave observations in space have been proposed. Among these, the most promising detectors are based on space-borne interferometry. In particular, *LISA* (Laser Interferometer Space Antenna) (Danzmann et al. 1993; Bender et al. 1994) and *OMEGA* (Orbiting Medium Explorer for Gravitational Astrophysics) (Hellings et al. 1995) consist of six drag-free, laser-bearing spacecraft, launched in orbit around the Sun (*LISA*) or the Earth (*OMEGA*). The six spacecraft would be placed, in pairs, at the vertices of a triangle with 5×10^6 km sides for *LISA*, and 5 times smaller for *OMEGA*. At each corner, the two spacecraft are phase-locked through the exchange of a laser signal, replacing in this way the central mirror of an ordinary Michelson interferometer. Each of the two probes sends a laser beam to a probe at each of the other two equilateral points, where the tracking signal is transponded

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back by phase-locked lasers, and the returning beams are eventually interfered.

In order to keep the triangle constellation as stable as possible, elaborate orbits have been designed. In the *LISA* case, each spacecraft is orbiting a circle of radius 3×10^6 km over a period of 1 yr. The interferometer plane, at an inclination of 60° with respect to the ecliptic, is also rotating around the Sun with the same periodicity. In the *OMEGA* case, the orbital plane is almost coincident with the ecliptic, and the interferometer is rotating around itself with a period of 53.21 d.

The complicated motion is reflected in the time evolution of the response of the interferometer to a source located in a fixed position in the sky. We will investigate the behaviour of the antenna response in the presence of a generic motion, and apply our results to the specific motions of interest. As a side-product of our analysis, we can also examine a terrestrial interferometer, where the motion is simply related to the Earth's rotation around its axis, and study its antenna pattern as a function of the location and orientation on the Earth's surface.

In this section, we briefly recall the formalism describing the antenna response to a gravitatinal wave passing by, in the long-wavelength approximation (Thorne 1987; see also Dhurandhar & Tinto 1988).

First, we introduce the wave symmetric trace-free (STF) tensor

$$\boldsymbol{W} = h_+ \Re(\boldsymbol{m}\otimes\boldsymbol{m}) + h_\times \Im(\boldsymbol{m}\otimes\boldsymbol{m}), \tag{1}$$

where the (complex) vector $\boldsymbol{m}$ is defined in terms of the polarization vectors $\boldsymbol{e}_X$ and $\boldsymbol{e}_Y$ as

$$\boldsymbol{m} = \frac{1}{\sqrt{2}}(\boldsymbol{e}_X + \mathrm{i}\boldsymbol{e}_Y). \tag{2}$$

The tensor $\boldsymbol{W}$ represents the wave field as measured in the interferometer's proper rest-frame. Then we define the STF detector tensor as

$$\boldsymbol{D} = \boldsymbol{n}_1\otimes\boldsymbol{n}_1 - \boldsymbol{n}_2\otimes\boldsymbol{n}_2, \tag{3}$$

where $\boldsymbol{n}_i$ is the unit vector along the ith arm. The interferometer response is the scalar obtained from the contraction of the wave tensor $\boldsymbol{W}$ with the detector tensor $\boldsymbol{D}$,

$$R(t) = W_{ij}D^{ij} \equiv F_+ h_+ + F_\times h_\times . \tag{4}$$

The beam-pattern factors F_+ and $F_\times$ depend on the orientation of the antenna with respect to the propagation direction and polarization axes of the wave.

We can choose the reference frame as in Fig. 1, with the x axis of the (x,y,z) frame bisecting the interferometer's arms, so that the only non-null components of $\boldsymbol{D}$ in this reference frame are

$$D_{12} = D_{21} = \sin(2\Omega), \tag{5}$$

where 2Ω is the aperture angle. Therefore, in order to obtain $R(t)$, we just need the component W_{12} in this particular frame. Fig. 1 also shows the Euler angles θ, ϕ and ψ, which transform from the (x,y,z) frame of the interferometer to the wave reference frame (X,Y,Z). The latter is defined with the Z axis opposite to the propagation direction, and the X and Y axes along $\boldsymbol{e}_X$ and $\boldsymbol{e}_Y$, respectively.

It is easy to find, for the + polarization,

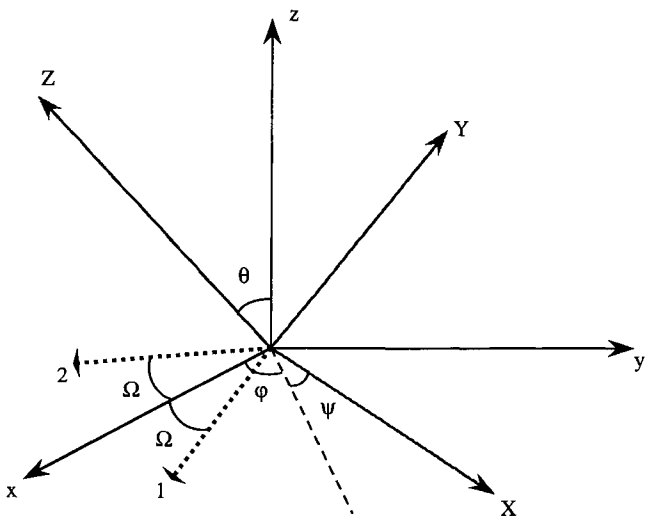


Figure 1. The geometry of the interferometer. Note that the usual spherical polar coordinates of the source's position are $(\theta, \phi - \pi/2)$.

$$F_+ = \sin(2\Omega)[\cos\theta\cos(2\phi)\sin(2\psi) + \frac{1}{2}(1+\cos^2\theta)\sin(2\phi)\cos(2\psi)]. \tag{6}$$

The beam-pattern factor $F_\times$ is obtained from equation (6) with the substitution $\psi \to \psi + \pi/4$, a well-known polarization property of gravitational waves. However, when averages over the polarization angle ψ are considered, we can assume, without loss of generality, $h^+ = h^\times = h$. The quantity of interest is thus the polarization-averaged *power pattern* of the antenna

$$P \equiv \left\langle \left(\frac{R}{\sin(2\Omega)h} \right)^2 \right\rangle_\psi , \tag{7}$$

which for the interferometer in Fig. 1 reads

$$P(\theta,\phi) = \frac{1}{8}[1 + \cos^4\theta + 6\cos^2\theta - \sin^4\theta\cos(4\phi)]. \tag{8}$$

For future reference, note that this definition of the antenna pattern is not normalized to unity, the average of $P(\theta,\phi)$ over the whole sky being 2/5. A plot of $P(\theta,\phi)$ in polar coordinates is shown in Fig. 5(a).

Equation (8) gives the instantaneous power pattern for a wave impinging from the direction (θ,ϕ) in the reference frame of the interferometer. If the detector is moving with respect to the source, then, apart from Doppler effects considerations, all we need to do is simply replace θ and ϕ by the appropriate functions $\theta(t)$ and $\phi(t)$. For example, if the antenna is rotating around its vertical axis with angular velocity ω (Fig. 2), then we can obtain the antenna pattern at any time t, from equation (8), with the substitution

$$\phi \to \phi - \xi_0 - \omega t, \tag{9}$$

where ξ_0 is some initial angle. As a matter of fact, this very simple case describes, to a good approximation, the time

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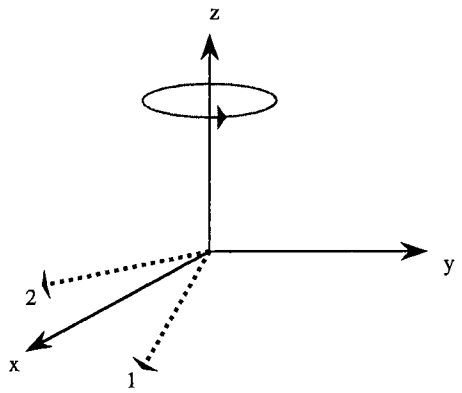


Figure 2. *OMEGA*: a space-borne detector rotating around its vertical axis. Although the actual *OMEGA*'s motion is better described with the formalism introduced in Section 3, this simple example can provide a good approximation to it.

evolution of *OMEGA* (Hellings et al. 1995). As mentioned in the introduction, the idea of *OMEGA* is essentially similar to that of *LISA*, except that the six spacecraft are launched into a circular Earth orbit, beyond the Moon orbit. The triangle has now 10^6-km sides, and it is rotating around itself with a period of ~ 53 d.

However, to mask the motion of the antenna with the apparent motion of the source is not always convenient, especially when dealing with a large number of sources, or when the motion is very complicated. In this paper we shall introduce a more useful representation, where every quantity is referred to a fixed reference frame, so that the polar coordinates θ and ϕ of the source remain constant, and the antenna response depends on time through the actual motion of the interferometer.

2 ANTENNA PATTERN FOR A GENERIC MOTION

We now introduce an arbitrary reference frame (x', y', z'), with the only requirement being that it be stationary with respect to fixed stars.[1] To be more explicit, when dealing with a space-borne interferometer, we can adopt the ecliptic coordinate system. In the last section we will also consider a terrestrial interferometer, which is most easily described in an equatorial frame.

The full description of the antenna response requires six Euler angles, defined as in Fig. 3. The orthogonal transformation from the (X, Y, Z) frame of the wave to the fixed frame (x', y', z') is given by the orthogonal matrix

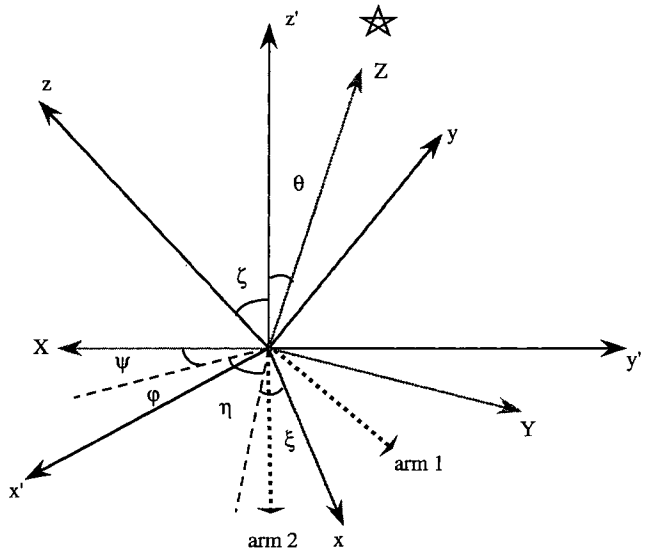


Figure 3. The relation between the frame of the wave (X, Y, Z), that of the detector (x, y, z) and the fixed one (x', y', z').

The matrix **B** which transforms from (x, y, z) to (x', y', z') is analogous to the matrix **A**, with the Euler angles θ, ϕ and ψ replaced by the corresponding ones ζ, η and ξ. Thus the complete transformation from the frame of the wave to that of the detector is given by $\mathbf{B}^{\mathrm{T}} \cdot \mathbf{A}$. Actually, as we may expect from Fig. 3, the angles ϕ and η appear in our results only in the combination $\delta \equiv \phi - \eta$.

After a rather lengthy calculation, one ends up with the following expressions for the beam-pattern factors

$$F_+ = \sin(2\Omega)[A\cos(2\xi)\cos(2\psi) + B\cos(2\xi)\sin(2\psi) + C\sin(2\xi)\cos(2\psi) + D\sin(2\xi)\sin(2\psi)], \quad (11a)$$

$$F_\times = \sin(2\Omega)[B\cos(2\xi)\cos(2\psi) - A\cos(2\xi)\sin(2\psi) + D\sin(2\xi)\cos(2\psi) - C\sin(2\xi)\sin(2\psi)]. \quad (11b)$$

The coefficients A, B, C and D in equations (11) depend only on the angles ζ, θ and δ. They are explicitly given in Appendix A.

We recall that the power pattern P is obtained by squaring, and averaging over the polarization angle ψ, the interferometer response. From equations (4), (7) and (11) we eventually obtain

$$\mathbf{A} = \begin{pmatrix} \cos\phi\cos\psi - \cos\theta\sin\phi\sin\psi & -(\cos\phi\sin\psi + \cos\theta\sin\phi\cos\psi) & \sin\theta\sin\phi \\ \sin\phi\sin\psi + \cos\theta\cos\phi\sin\psi & -\sin\phi\sin\psi + \cos\theta\cos\phi\cos\psi & -\sin\theta\cos\phi \\ \sin\theta\sin\psi & \sin\theta\cos\psi & \cos\theta \end{pmatrix}. \quad (10)$$

[1]Here and in the following, when we say 'stationary' (or 'fixed') we mean stationary over the characteristic time-scale of the motion of the detector.

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$$P=\sum_{n=0}^{4}\{[\lambda_n+\mu_n\cos(4\xi)]\cos(n\delta)+\sigma_n\sin(4\xi)\sin(n\delta)\}, \tag{12}$$

where now the coefficients λ_n, μ_n and σ_n depend only on θ and ζ. These coefficients, given in Appendix B, are quite complicated trigonometric polynomials of their arguments. None the less, equation (12) turns out to be very useful in practice. In fact, for the planned detectors considered in the present paper, the angle ζ is constant, and thus the only possible time evolution is related to the sinusoidal functions of ξ and δ, which appear explicitly in equation (12). Before analysing in more detail the proposed space-borne interferometers, we consider a trivial application of equation (12).

2.1 A simple example

As a first test of equation (12) we can consider, as we did at the end of Section 1, an interferometer which is rotating around its z axis (see Fig. 2), so that its trivial motion is described by

$$\zeta=\eta=0, \qquad \xi=\xi_0+\omega t. \tag{13}$$

From equation (13), and equations (B1)–(B15) of Appendix B, we find that the only non-null coefficients in equation (12) are

$$\lambda_0=\frac{1}{8}(1+\cos^4\theta+6\cos^2\theta), \tag{14}$$

$$\mu_4=\sigma_4=-\frac{1}{8}\sin_4\theta, \tag{15}$$

so that

$$P=\frac{1}{8}\{1+\cos^4\theta+6\cos^2\theta-\sin^4\theta\cos[4(\phi-\xi_0-\omega t)]\}. \tag{16}$$

As pointed out in Section 1, this result can be obtained much more easily directly from equation (8), with the substitution (9). Equation (16) gives, to a good approximation, the antenna pattern of *OMEGA*, with $\omega\simeq 1.4\times10^{-6}\ \mathrm{s}^{-1}$.

3 SPACE INTERFEROMETERS: CONICAL MOTION

In the previous section, we have considered the antenna power pattern associated with an unspecified motion of the detector. We will now focus our attention on the case of a space interferometer, which presents, independently of the particular project under investigation, some very general and interesting properties.

Inserting the space interferometer in its orbit and keeping the interferometer configuration stable over the emission lifetime – at least two orbital periods – is a very demanding navigation task, due principally to the perturbations from the Earth and the other planets. For instance, one of the consequences of the complicated orbit is the fact that we cannot maintain equal distances between the probes. In a recent paper (Giampieri et al. 1996) we modelled the noise that is introduced into the differenced data because of the unequal arms, and showed that the final accuracy of the interferometer is not compromised. Another example of the problems we may face in a space-borne interferometer is that, due to the Earth disturbances, high Doppler rates would result. Hellings et al. (1996) described a laser transmitter and receiver hardware system that provides the required readout accuracy and implements a self-correction procedure for the on-board frequency standard used for laser phase measurement.

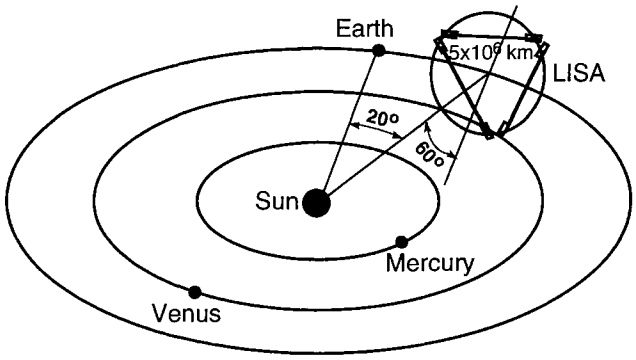


Figure 4. *LISA*: a space-borne interferometer in orbit around the Sun.

We will now consider the orbit of the interferometer, and discuss its implication on the antenna response to a wave coming from a given direction in the sky. In particular, as we mentioned in the introduction, *LISA* (Danzmann et al. 1993; Bender et al. 1994) will orbit the Sun at the Earth's distance, as far behind the Earth as possible. The plane containing the six probes, during its orbit, will remain always tangential to the surface of a cone of 60° aperture, and the detector itself will rotate in this plane with same periodicity (1 yr) but in the opposite direction. Fig. 4 shows the *LISA* configuration.

In this section we will consider a *LISA*-like motion, characterized by a generic cone aperture. In other words, the motion of each of the three interferometers is assumed to be described by[2]

$$\zeta=\text{constant}, \tag{17a}$$

$$\eta=\eta_0+\omega t \qquad (\Rightarrow\delta=\phi-\eta_0-\omega t), \tag{17b}$$

$$\xi=\xi_0-\omega t. \tag{17c}$$

Note that η and ξ are counter-rotating. Note also that the three interferometers formed by the triangular configuration have initial values ξ_0 which differ from reach other by 120°, whereas η_0 is the same for all of them. For the sake of conciseness, since we are concerned here with the antenna power pattern, we will consider only one of the interferometers, leaving the possibility of expoiting the polarization sensitivity to future work.

[2]We could eliminate one of the two initial conditions ξ_0 or η_0 by simply rescaling the time, taking for instance the origin of time at the passage through the line of nodes ($\xi_0=0$) or through the vernal equinox ($\eta_0=0$).

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The proposed *LISA* orbit has $\zeta=60°$, a critical value for the stability of the triangular configuration. Since the behaviour of the antenna, in terms of sky coverage, directionality, etc., is very sensitive to the inclination, we will keep ζ as a free parameter throughout this paper, and refer to equations (17) as describing a 'conical' motion. For a given ζ, the coefficients λ_n, μ_n and σ_n are now functions of θ. They are explicitly given in Appendix C for the *LISA* case.

In order to determine the sky's coverage during the detector's lifetime, we need to consider the time average of the antenna pattern P over one orbital period, which gives

$$\langle P\rangle_T=\lambda_0+\frac{1}{2}(\mu_4+\sigma_4)\cos[4(\phi-\eta_0-\xi_0)]. \tag{18}$$

Fig. 5 shows a plot of $\langle P\rangle_T$ in polar coordinates for various values of the aperture of the cone ζ. Note that $\zeta=0$ corresponds to an interferometer fixed in space; since we are taking ξ and η as counter-rotating, the interferometer does not rotate at all.

We know from equation (8) that an interferometer fixed in space can never detect waves impinging from four specific null-directions, given by $\theta=\pi/2$, and $\phi=k\pi/2$ $(k=0,\ldots,3)$. We can now analyse what happens to these null-directions in the generic conical case, focusing our attention to the ecliptic plane $\theta=\pi/2$. It is easy to show, from the behaviour of the functions λ_0 and $|\mu_4+\sigma_4|/2$ (see Fig. 6), that there is a tendency for the null-directions on the ecliptic plane to remain visible, although the magnitude of this effect is strongly affected by the value of ζ: for $\zeta=0$ the null-directions are obviously completely preserved, while for $\zeta=\pi/2$ the ϕ-dependence is very poor. *LISA* is much closer to the latter case, and actually its $\langle P\rangle_T$ is almost independent of θ as well, as we shall see later.

Note that, if ξ and η were co-rotating, instead of counter-rotating as in *LISA*, then in the last term of equation (18) we would have to make the substitution

$$\frac{1}{2}(\mu_4+\sigma_4)\rightarrow\frac{1}{2}(\mu_4-\sigma_4). \tag{19}$$

which means that the ϕ-term would be even smaller, in absolute value, compared with λ_0.

One might infer from Fig. 5(d) that, if the inclination is close to $\zeta=\pi/2$, then the zeros of the antenna pattern can be found in the direction orthogonal to the ecliptic plane. However, this is not the case, since $\theta=0$ implies that P is still given by equation (8), with θ and ϕ replaced, respectively, by ζ and ξ. Thus, even for $\zeta=\pi/2$ the time average is non-zero at the poles.

Another important issue related to equation (18) is the time-averaged antenna directionality. As we can predict from Fig. 5, directionality is strongly dependent on the angle ζ. To make this statement more precise, let us consider the rms deviation from isotropy, defined as

$$\Delta\equiv\left[\frac{175}{48\pi}\int_{4\pi}(\langle P\rangle_T-2/5)^2\,\mathrm{d}\Omega\right]^{1/2}. \tag{20}$$

The normalization factor in front of equation (20) is chosen in such a way that Δ is normalized to one for a fixed interferometer. For a conical motion, inserting equation (18) in equation (20) we easily obtain $\Delta=\Delta(\xi)$, shown in Fig. 7. Explicitly, we find

$$\Delta(\xi)=\frac{1}{288}(19779+120\cos\zeta-118380\cos^2\zeta+840\cos^3\zeta+180690\cos^4\zeta+840\cos^5\zeta-3180\cos^6\zeta+120\cos^7\zeta+2115\cos^8\zeta)^{1/2}. \tag{21}$$

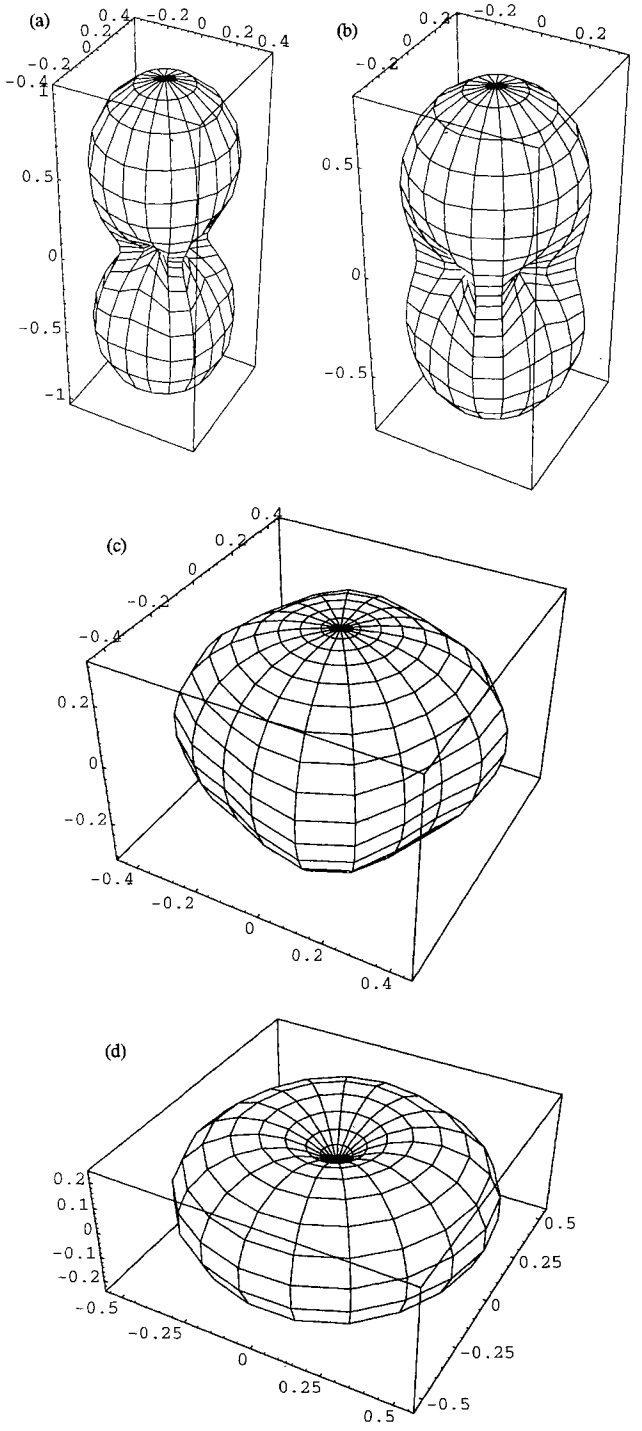


Figure 5. The antenna pattern, averaged over time and polarization for various values of the inclination angle ζ: (a) $\zeta=0$, (b) $\zeta=30°$, (c) $\zeta=60°$, (d) $\zeta=90°$. *LISA* corresponds to case (c).

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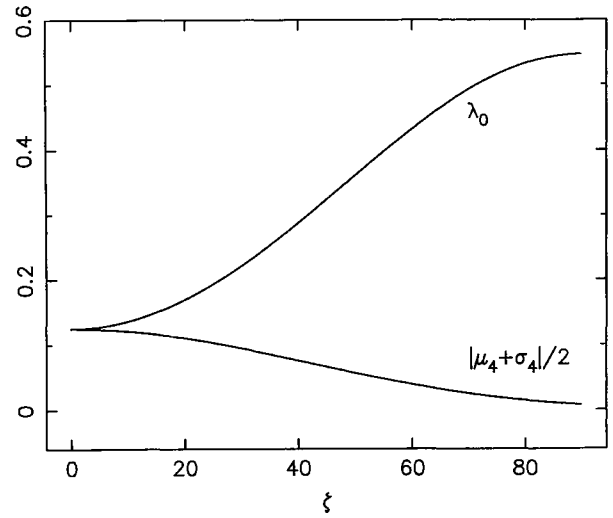


Figure 6. The ϕ-dependence on the ecliptic plane, as a function of the angle ζ. The bigger is the difference between λ_0 and $|\mu_4+\sigma_4|/2$; the smaller is the dependence on the direction ϕ in the averaged antenna pattern.

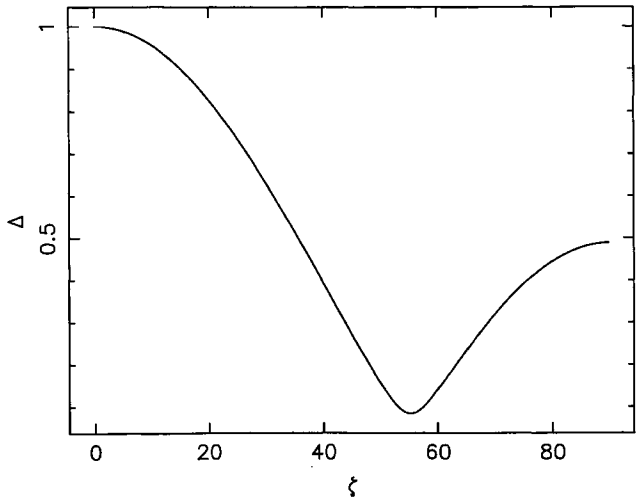


Figure 7. The rms deviation from isotropy of the time-averaged antenna pattern, as a function of ζ. The minimum of Δ corresponds to the maximum attainable isotropic response. This minimum occurs at $\zeta \simeq 55°$, for which $\Delta \simeq 0.08$. For *LISA*, $\Delta(60°) \simeq 0.14$.

Thus the antenna pattern is distributed more and more isotropically as we increase ζ from $\zeta = 0$. After we reach a minimum at $\zeta = 55°$, the directionality of the antenna starts increasing again.

Therefore we can conclude that, in the conical case, it is impossible to get a perfectly isotropic response, i.e., we never get $\Delta = 0$ (see also the above discussion about the ϕ-dependence). However, we can get very close to this ideal situation if we choose ζ appropriately. Remarkably, *LISA*'s inclination is very close to the optimal value $\zeta \simeq 55°$, and gives $\Delta(60°) \simeq 0.14$. This means that *LISA*, during its lifetime, will cover the whole sky in an approximately uniform manner. Of course, in some circumstances, directionality needs to be preserved. For instance, one may want to disentangle the isotropic component of the stochastic background from the anisotropic contribution of the Galactic binaries (Giampieri & Polnarev 1997). Directionality can always be preserved integrating over a shorter period. Fig. 8 shows *LISA*'s antenna pattern averaged over 3 and 6 months, respectively.

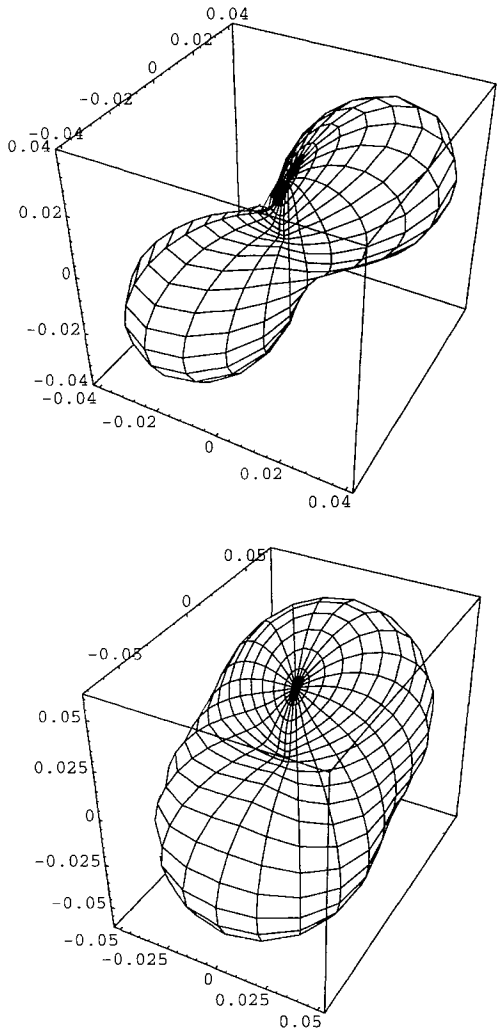


Figure 8. *LISA*'s antenna pattern averaged over (a) 3 months and (b) 6 months. The initial conditions are $\xi_0 = \eta_0 = 0$.

Finally, we want to consider the interferometer's responsiveness to the Galactic plane, given that most of the strongest sources will lie in this plane. In our coordinate system, the Galactic plane is characterized by

$$\theta(\phi) = \arctan(\alpha \cos\phi + \beta \sin\phi)^{-1}, \tag{22}$$

where $\alpha \simeq 1.75$, and $\beta \simeq 4 \times 10^{-4}$. Assuming, for simplicity, that the sources are distributed isotropically in the plane, the event rate of disc's sources is proportional to the average area of the interaction of this plane with the antenna pattern, equal to

$$G = \frac{1}{2T} \int_0^{2\pi} \int_0^T \{P[\theta(\phi), \phi]\}^2 \, \mathrm{d}t \, \mathrm{d}\phi, \tag{23}$$

where $\theta(\phi)$ is the function given in equation (22). This area depends on the details of the motion of the detector, in our

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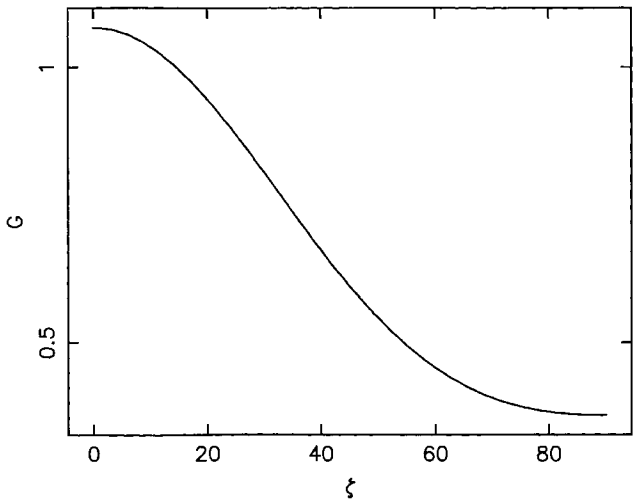


Figure 9. The response of a space interferometer to sources on the Galactic disc, as a function of the angle ζ. For simplicity, a small dependence on the detector's initial orientation, more precisely on $\eta_0 + \xi_0$, has been neglected.

case on ζ, ξ_0 and η_0. Fig. 9 shows the quantity G as a function of ζ, since the dependence on ξ_0 and η_0 can be neglected in a first approximation. For *LISA*, we find the small value $G \sim 0.46$, the exact value depending on the initial conditions $\eta_0 + \xi_0$. We conclude that *LISA* is not particularly sensitive to the Galactic disc, due to the fairly large inclination to the ecliptic of both the plane of the detector and the Galaxy. According to this crude analysis, we can expect that *OMEGA*, with a smaller, almost negligible, inclination, would increase its chances of observing a signal from the disc by roughly a factor of 2. Note, however, that previous calculations (Evans, Iben & Smarr 1987; Hils, Bender & Webbink 1990) have shown that gravitational radiation from Galactic binaries in the disc is comparable to that coming from the local region ($r < 200$ pc). Since the latter is isotropically distributed, the time-varying signal from the disc contributes only a fraction of the Galactic binaries' stochastic background. None the less, this small component could make the stochastic signal distinguishable from the detector's noise (Giampieri & Polnarev 1997). In a future paper we will investigate the amplitude modulations introduced by the antenna motion in the confusion noise generated by different populations of Galactic binary systems, and describe how to exploit this effect in order to detect the signal and to obtain information about the distribution of sources in the Galaxy.

4 EARTH-BASED INTERFEROMETERS

As an additional application of equation (12), let us consider a terrestrial interferometer. In this case the most convenient choice for the 'fixed' reference frame is the equatorial one, with the z' axis directed toward the North Pole, and the x' axis toward the vernal equinox. In this frame, the motion of the interferometer becomes similar to the conical case previously analysed, except that now the detector cannot rotate on itself, of course. In other words, we have

$$\zeta = \pi/2 - \ell, \tag{24a}$$

$$\eta = \eta_0 + \omega t, \tag{24b}$$

$$\xi = \iota, \tag{24c}$$

where ℓ is the terrestrial latitude, ω is the Earth's angular velocity of rotation, and $\iota \in [-\pi/4, +\pi/4]$ is the angle between the arms' bisector and the local parallel.

For example, let us consider the average of the antenna pattern P over the time of arrival of the signal. In the terrestrial case, as opposed to the conical case considered in Section 3, the interferometer cannot rotate around its vertical axis, and therefore averaging over time or ϕ gives the same result, namely

$$\langle P \rangle_T = \lambda_0(\theta, \ell) + \mu_0(\theta, \ell) \cos(4\iota). \tag{25}$$

For any specific value of θ, for example, $\theta = 102°$, corresponding to the direction of the centre of the Virgo cluster, equation (25) gives the square of the rms power as a function of the antenna's position and orientation on the Earth's surface, a quantity already numerically studied by Schutz & Tinto (1987).

Instead of fixing θ, we could try to answer the question: is there any particular location and orientation for which the antenna pattern, averaged over one day, is isotropic? The answer turns out to be affirmative, and the isotropic antenna is characterized by

$$\ell_{\rm is} = \arcsin\left(\pm\frac{1}{\sqrt{3}}\right) \simeq \pm 35°, \tag{26a}$$

$$\iota_{\rm is} = \frac{1}{4}\arccos\left(-\frac{1}{5}\right) \simeq \pm 25°. \tag{26b}$$

One can easily check, by inspection, that $\ell_{\rm is}$ and $\iota_{\rm is}$ produce $\langle P \rangle_T \equiv 2/5$ or, equivalently, $\Delta \equiv 0$. A detector located at latitude $\ell_{\rm is}$ and oriented by $\iota_{\rm is}$ maximizes the event rate of an isotropic population of sources.

In the event that the detector's position has already been chosen, one can still make use of equation (25) in order to find the optimal orientation ι_* which gives the least directional antenna pattern at that latitude. At each latitude ℓ, we define as optimal that orientation ι_* that minimizes the quantity Δ. Fig. 10 shows $\iota_* = \iota_*(\ell)$ and the corresponding minimum $\Delta_* \equiv \Delta(\ell, \iota_*)$.

We stress that, as one may actually expect, Δ depends much more strongly on ℓ than ι, and in particular the antenna becomes rapidly anisotropic as we move away from $\ell_{\rm is}$, no matter how optimally we try to choose ι. Moreover, we can foresee several terrestrial interferometers to be operative in the near future, so that the sky's coverage of a single antenna is not really an issue as critical as in the space-borne case previously discussed.

5 SINUSOIDAL WAVES FROM BINARIES: FOURIER ANALYSIS

It is generally assumed that Galactic and extragalactic binary systems are the most promising sources of gravitational waves for detectors based on laser interferometry. In fact, waves from a binary star, including the effect of eccentricity, orbital inclination, and also post-Newtonian cor-

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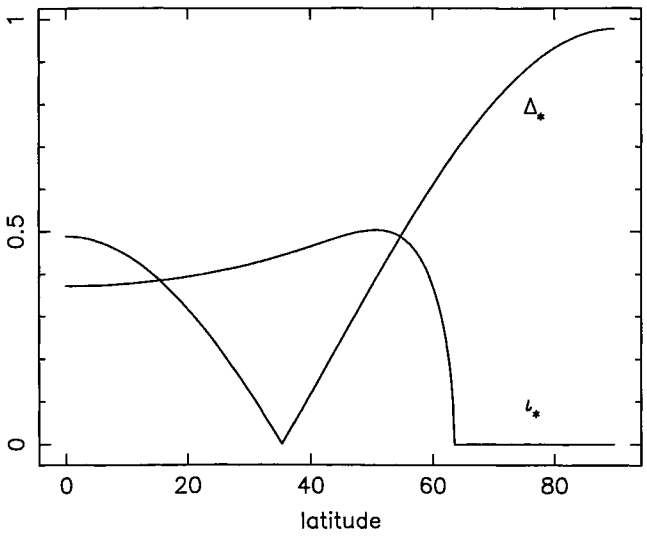


Figure 10. The minimum rms deviation from isotropy Δ_* attainable at any given latitude. The corresponding optimal orientation ι_* is also shown.

rections, have long been studied, and are now well understood. In particular, the sensitivity of the planned space interferometers should allow the detection of waves from several known Galactic binary stars. In the *LISA* and *OMEGA* frequency band, the strongest among these sources are presumably interacting white dwarf binaries (IWDBs) (Evans et l. 1987; Hils et al. 1990). Table 1 contains the available data for five IWDBs, including the amplitude and frequency of the expected gravitational waves. We have applied our results to these objects, and described *LISA*'s response to the waves originating from them. Fig. 11 shows the beam-pattern factors F_+ and $F_\times$ for the five IWDBs in Table 1, as seen from *LISA* over 1 yr. Fig. 11 also shows the analogous quantities for a sinusoidal signal, of unspecified amplitude and frequency, coming from the Galactic Centre.

We will now consider the effect of the motion in the frequency domain, for both the space-borne and the terrestrial cases. We define the Fourier series as usual:

$$P(t)=a_0+\sum_{k=1}^{\infty}\left[a_k\cos\left(\frac{k\pi t}{L}\right)+b_k\sin\left(\frac{k\pi t}{L}\right)\right], \tag{27}$$

where $T=2L$ is the orbital period, and

$$a_0=\frac{1}{2L}\int_{-L}^{+L}P(t)\,\mathrm{d}t, \tag{28}$$

$$a_k=\frac{1}{L}\int_{-L}^{+L}P(t)\cos\left(\frac{k\pi t}{L}\right)\mathrm{d}t, \tag{29}$$

$$b_k=\frac{1}{L}\int_{-L}^{+L}P(t)\sin\left(\frac{k\pi t}{L}\right)\mathrm{d}t. \tag{30}$$

For the conical case analysed in Section 3, using equations (B1)–(B14) of Appendix B, one finds that the only non-zero Fourier coefficients are given by ($\delta_0\equiv\phi-\eta_0$, $k=1,\ldots,8$):

$$a_0=\lambda_0+\frac{1}{2}(\mu_4+\sigma_4)\cos[4(\delta_0-\xi_0)], \tag{31}$$

$$\begin{aligned}a_k=\mu_0\cos(4\xi_0)\,\delta_{k4}+\sum_{n=1}^{4}\Big[&\lambda_n\cos(n\delta_0)\,\delta_{nk}\\&+\frac{1}{2}(\mu_n+\sigma_n)\cos(n\delta_0-4\xi_0)\,\delta_{k+n,4}\\&+\frac{1}{2}(\mu_n-\sigma_n)\cos(n\delta_0+4\xi_0)\,\delta_{k-n,4}\Big],\end{aligned} \tag{32}$$

$$\begin{aligned}b_k=\mu_0\sin(4\xi_0)\,\delta_{k4}+\sum_{n=1}^{4}\Big[&\lambda_n\sin(n\delta_0)\,\delta_{nk}\\&-\frac{1}{2}(\mu_n+\sigma_n)\sin(n\delta_0-4\xi_0)\,\delta_{k+n,4}\\&+\frac{1}{2}(\mu_n-\sigma_n)\sin(n\delta_0+4\xi_0)\,\delta_{k-n,4}\Big].\end{aligned} \tag{33}$$

The analogous calculation for the terrestrial case, described by equations (24), gives the following nine coefficients ($k=1,\ldots,4$):

$$a_0=\lambda_0+\mu_0\cos(4\iota) \tag{34}$$

$$a_k=[\lambda_k+\mu_k\cos(4\iota)]\cos(k\delta_0)+\sigma_k\sin(4\iota)\sin(k\delta_0), \tag{35}$$

$$b_k=[\lambda_k+\mu_k\cos(4\iota)]\sin(k\delta_0)-\sigma_k\sin(4\iota)\cos(k\delta_0). \tag{36}$$

Table 1. Data for five known IWDBs. The first column is the name, and the second and third columns are, respectively, the angles θ and ϕ in the ecliptic coordinate system. The last two columns give, respectively, the predicted amplitude and frequency of the gravitational waves.

Name	θ (degrees)	ϕ (degrees)	GW Amplitude (10^{-22})	GW Frequency (10^{-3} Hz)
AM CVn	52.56	260.38	5.27	1.94
CR Boo	72.10	292.27	2.82	1.34
V803 Cen	120.31	306.17	0.89	1.24
CP Eri	120.83	151.77	4.02	1.16
GP Com	67.00	277.73	1.77	0.72

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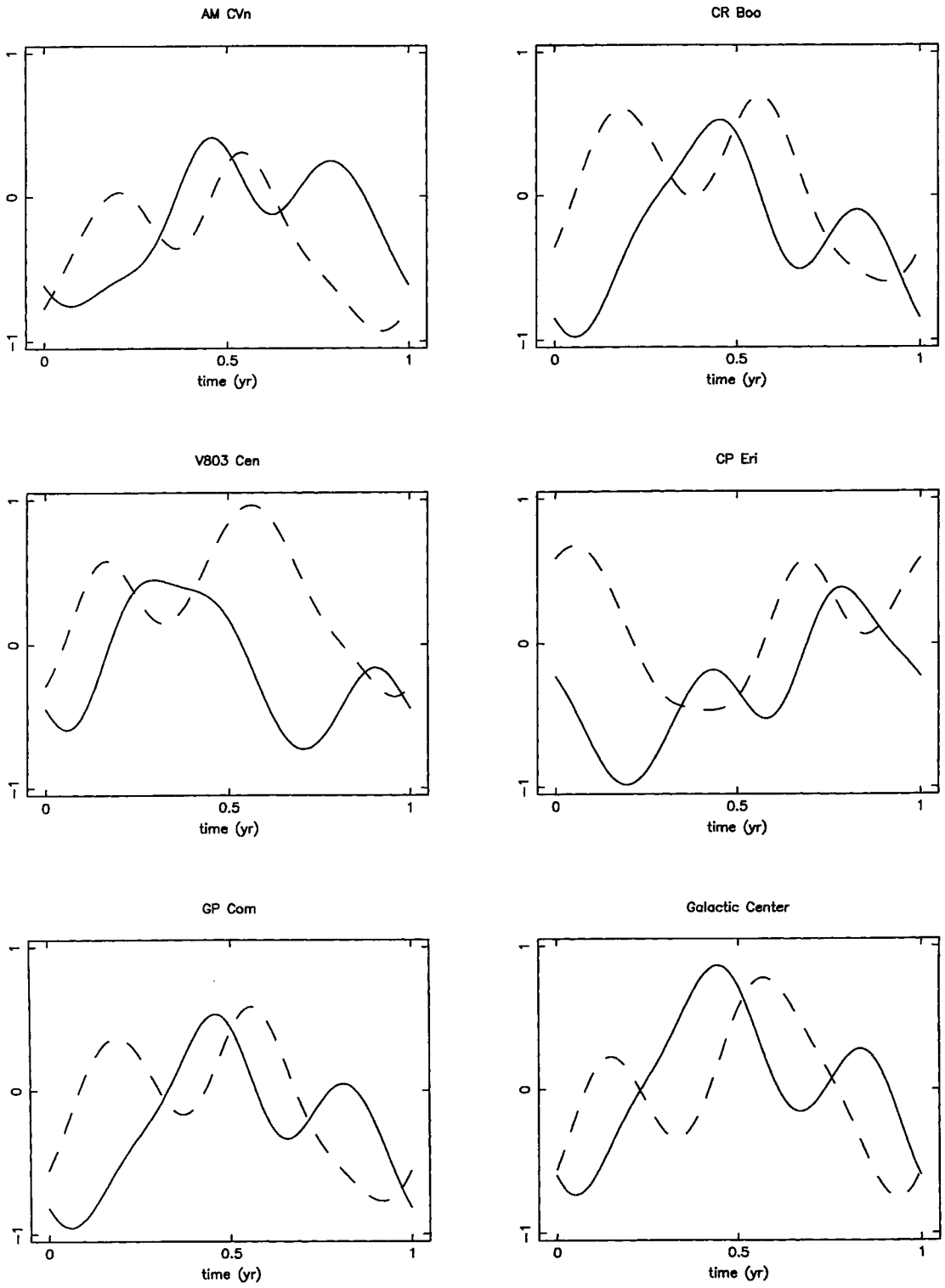


Figure 11. The *LISA* normalized responses to the five IWDB systems listed in Table 1, and to a hypothetical source in the Galactic Centre, over one year of observations. The two curves give the beam-pattern factors F_+ (solid line) and $F_\times$ (dashed line). We have assumed the initial conditions $\xi_0=\eta_0=0$, and a polarization angle $\psi=2$. Horizontal axis is time in years.

In general, given the initial detector's position, these Fourier coefficients depend on the coordinates of the source, θ (through λ_k, μ_k and σ_k) and ϕ (through δ_0). If the source location is known, then one can look for these spectral lines as a convincing signature about the gravitational origin of the signal. When the coordinates of the source are unknown, however, one has to deal with the complication arising from the Doppler effect (Schutz 1991). The motion of the detector, besides the amplitude modulation described in the present work, also introduces a location-dependent

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phase modulation, in the form of a Doppler broadening of the sinusoidal signal. In the case of *LISA*, the magnitude of this effect, over a period $T=1$ yr, is

$$\frac{\Delta f}{f}\simeq 2\times 10^{-4}, \tag{37}$$

so that, in the spectral region below 10^{-2} Hz, we do not have any hope of finding the aforementioned lines, separated from each other by only $1/T\simeq 3\times 10^{-8}$ Hz. For a terrestrial interferometer, the situation is analogous, but complicated by the simultaneous effects of the diurnal and annual motion of the Earth, and also by the Earth–Moon interaction.

In conclusion, for long enough observations, we need special techniques to compensate for the frequency spread over several frequency-resolution bins, and eventually to recover the amplitude modulation described in this paper. Several different strategies can be adopted to overcome this problem, although none of them is completely satisfactory, due to the large amount of computation involved; see Schutz (1991) for more details. In any case, the amplitude modulation can be exploited for an independent measurement of the location of the source, and, in addition, to obtain the polarization of the wave.

6 CONCLUSIONS

In the present work we have considered a gravitational wave interferometer, in motion with respect to fixed stars, and studied the resulting amplitude modulation of a long-duration continuous signal. The general results are presented in Section 2, where the instantaneous beam-pattern factors (equations 11) and the polarization-averaged antenna power pattern (equation 12) are given as functions of time, for a generic motion.

Next, two particular cases have been analysed: (1) the probable orbit of a space-borne interferometer, with particular emphasis on *LISA*, and (2) the motion of a ground-based interferometer.

For what concerns *LISA*, we have shown that its peculiar motion makes the time-averaged antenna pattern practically isotropic, thus providing a uniform coverage of the whole sky over the period of 1 yr. For shorter integration periods directionality is mostly preserved, and can be exploited where necessary, for example, in the search for a Galactic binaries background. However, when we focused on the Galactic disc, we found that the average antenna response is far from optimal, due to the relative orientation of the ecliptic and the Galactic plane itself. We stress that these results are not conclusive, since we have neglected the anisotropy in the distribution of the sources with respect to the Sun, due to the fact that we are located near the edge of the disc. In this respect, additional work is needed.

In the terrestrial case, thanks to the probable redundancy of future gravitational wave observatories, the discussion about the sky coverage of a single antenna is not so critical. However, we found that there are particular positions on the Earth's surface, given in equations (26), which render the time-averaged antenna response perfectly isotropic. For what concerns the Galactic plane, since the latter makes with the equatorial plane approximately the same angle it makes with the ecliptic ($\sim 60°$), the result is analogous to the conical case, with the angle ζ interpreted as $90-\ell$ in Fig. 9. In other words, the response to the Galactic plane increases as we move the interferometer from the equator toward the poles, with a minor role played by the orientation angle ι.

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APPENDIX A

In this appendix we give the coefficients A, B, C and D which enter in the expressions of F_+ and $F_\times$, (equations 11). They, in turn, can be expressed in terms of intermediate quantities α_1, α_2, β and β_2 as follows:

$$A=\alpha_1\beta_1+\alpha_2\beta_2, \tag{A1}$$

$$B=\alpha_1\beta_2-\alpha_2\beta_1, \tag{A2}$$

$$C=\frac{1}{2}[\beta_1^2-\beta_2^2-\alpha_1^2+\alpha_2^2], \tag{A3}$$

$$D=\alpha_1\alpha_2+\beta_1\beta_2, \tag{A4}$$

where

$$\alpha_1=\cos\delta,$$

$$\alpha_2=\cos\theta\sin\delta,$$

$$\beta_1=\cos\zeta\sin\delta,$$

$$\beta_2=\cos\zeta\cos\theta\cos\delta+\sin\zeta\sin\theta.$$

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APPENDIX B

In this appendix we give the coefficients λ_n, μ_n and σ_n $(n=0,\ldots,4)$, defined in equation (12), as functions of the angles θ and ζ:

$$\lambda_0=\frac{1}{64}(35+35\cos^4\theta\cos^4\zeta-30\cos^2\theta+3\cos^4\zeta+108\cos^2\zeta\cos^2\theta-30\cos^2\zeta\cos^4\theta-30\cos^4\zeta\cos^2\theta-30\cos^2\zeta+3\cos^4\theta), \quad \text{(B1)}$$

$$\lambda_1=\frac{1}{32}\sin(2\zeta)\sin(2\theta)(15-3\cos^2\theta-3\cos^2\zeta+7\cos^2\zeta\cos^2\theta), \quad \text{(B2)}$$

$$\lambda_2=\frac{1}{16}(7+7\cos^4\theta\cos^4\zeta-8\cos^2\theta+\cos^4\zeta+16\cos^2\zeta\cos^2\theta-8\cos^2\zeta\cos^4\theta-8\cos^4\zeta\cos^2\theta-8\cos^2\zeta+\cos^4\theta), \quad \text{(B3)}$$

$$\lambda_3=\frac{1}{8}\sin^3\zeta\cos\zeta\sin^3\theta\cos\theta, \quad \text{(B4)}$$

$$\lambda_4=\frac{1}{64}\sin^4\zeta\sin^4\theta; \quad \text{(B5)}$$

$$\mu_0=-\frac{1}{64}\sin^4\zeta(3-30\cos^2\theta+35\cos^4\theta), \quad \text{(B6)}$$

$$\mu_1=-\frac{1}{16}\sin(2\theta)\cos\zeta\sin^3\zeta(3-7\cos^2\theta), \quad \text{(B7)}$$

$$\mu_2=\frac{1}{16}\sin^2\zeta\sin^2\theta(1+\cos^2\zeta)(1-7\cos^2\theta), \quad \text{(B8)}$$

$$\mu_3=\frac{1}{16}\sin(2\zeta)\sin^3\theta\cos\theta(3+\cos^2\zeta), \quad \text{(B9)}$$

$$\mu_4=-\frac{1}{64}\sin^4\theta(1+\cos^4\zeta+6\cos^2\zeta); \quad \text{(B10)}$$

$$\sigma_0=0, \quad \text{(B11)}$$

$$\sigma_1=-\frac{1}{16}\sin(2\theta)\sin^3\zeta(3-7\cos^2\theta), \quad \text{(B12)}$$

$$\sigma_2=\frac{1}{8}\cos\zeta\sin^2\zeta\sin^2\theta(1-7\cos^2\theta), \quad \text{(B13)}$$

$$\sigma_3=\frac{1}{8}\sin\zeta\cos\theta\sin^3\theta(1+3\cos^2\zeta), \quad \text{(B14)}$$

$$\sigma_4=-\frac{1}{16}\sin^4\theta\cos\zeta(1+\cos^2\zeta). \quad \text{(B15)}$$

APPENDIX C

In this appendix we provide the coefficients λ_n, μ_n and σ_n $(n=0,\ldots,4)$ for *LISA*:

$$\lambda_0=\frac{1}{1024}(443-37\cos^4\theta-78\cos^2\theta), \quad \text{(C1)}$$

$$\lambda_1=\frac{\sqrt{3}}{256}\sin(2\theta)(57-5\cos^2\theta), \quad \text{(C2)}$$

$$\lambda_2=\frac{9}{256}(9-\cos^4\theta-8\cos^2\theta), \quad \text{(C3)}$$

$$\lambda_3=\frac{3\sqrt{3}}{128}\sin^3\theta\cos\theta, \quad \text{(C4)}$$

$$\lambda_4=\frac{9}{1024}\sin^4\theta; \quad \text{(C5)}$$

$$\mu_0=-\frac{9}{1024}(3+35\cos^4\theta-30\cos^2\theta), \quad \text{(C6)}$$

$$\mu_1=-\frac{3\sqrt{3}}{256}\sin(2\theta)(3-7\cos^2\theta), \quad \text{(C7)}$$

$$\mu_2=\frac{15}{256}\sin^2\theta(1-7\cos^2\theta), \quad \text{(C8)}$$

$$\mu_3=\frac{13\sqrt{3}}{128}\sin^3\theta\cos\theta, \quad \text{(C9)}$$

$$\mu_4=-\frac{41}{1024}\sin^4\theta; \quad \text{(C10)}$$

$$\sigma_0=0, \quad \text{(C11)}$$

$$\sigma_1=-\frac{3\sqrt{3}}{128}\sin(2\theta)(3-7\cos^2\theta), \quad \text{(C12)}$$

$$\sigma_2=\frac{3}{64}\sin^2\theta(1-7\cos^2\theta), \quad \text{(C13)}$$

$$\sigma_3=\frac{7\sqrt{3}}{64}\sin^3\theta\cos\theta, \quad \text{(C14)}$$

$$\sigma_4=-\frac{5}{128}\sin^4\theta. \quad \text{(C15)}$$

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